Practice Quiz 11A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 11

Exhibit a bijection with the positive cubes.

1. Let $C = \{n^3 : n \in \mathbf{Z}^+\}$. Define a function $f : \mathbf{Z}^+ \to C$ and prove that it is a bijection. What does this say about the cardinalities of C and $\mathbf{Z}^+$?

   Solution Set $f(n) = n^3$. Its value is in C for every positive integer n. If $f(m) = f(n)$, then $m^3 = n^3$. Cubing is strictly increasing on positive integers, so $m = n$; thus f is one-to-one. For any $c \in C$, the definition of C supplies a positive integer n with $c = n^3 = f(n)$, so f is onto C. It is a bijection, and therefore $C \sim \mathbf{Z}^+$. ∎

Solutions prepared using GPT-6 in the Codex harness.